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A model for association in bivariate life tables and its application in epidemiological studies of familial tendency in chronic disease incidence

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Summary

The application of Cox's (1972) regression model for censored survival data to epidemiological studies of chronic disease incidence is discussed. A related model for association in bivariate survivorship time distributions is proposed for the analysis of familial tendency in disease incidence. The possible extension of the model to general multivariate survivorship distributions is indicated.

Some key words: Censored data; Epidemiological applications; Multivariate life table; Reliability theory.

1. Introduction

This paper is concerned with a problem in the analysis of epidemiological studies of chronic disease incidence. In contrast with problems in the epidemiology of infectious disease, such analysis usually assumes that incidence of disease in different individuals represents independent events, the occurrence of which is influenced by measurable factors describing individuals and their environment. However, in the study of familial tendency in chronic disease incidence, this assumption is called into question. Comparisons of parents and offspring and sibling comparisons investigate possible relationships between disease incidence in related individuals and such studies provide interesting analytical difficulties.

Here, this problem is treated as one of estimating association in multivariate life tables. In § 2 it is shown that epidemiological incidence studies may be regarded as being concerned primarily with the study of the distribution of the age at incidence and that Cox's (1972) regression model for the analysis of censored survival time data may readily be applied to incidence data and is closely related to more specifically epidemiological models. In later sections, a related model for bivariate life tables is developed and applied to the problem of demonstrating association in disease incidence in ordered pairs of individuals. The possible extension of the model into more dimensions is indicated.

2. The application of Cox's model in incidence studies

Epidemiological studies of disease incidence seek to relate an individual's risk of succumbing to a particular disease to certain measured characteristics of that individual. To this end, two types of study are commonly used:

(i) longitudinal studies of incidence in which individuals are followed forward in time from their admission to the study until they succumb to the disease studied or until their observation is cut short for other reasons, and

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(ii) case-control studies of incidence in which the characteristics of individuals who have succumbed to the disease, i.e. cases, are compared with those of individuals who have remained free of the disease, i.e. controls.

The greater part of this paper is concerned with longitudinal studies since, for the models considered, inference from case-control studies will be shown to be readily achievable with only a minor modification of the argument.

Longitudinal studies of incidence are similar in many ways to survival studies which set out to investigate the effects of characteristics of patients and of their treatment upon their subsequent survival. Both types of study are concerned with elucidating the factors influencing the 'failure' of individuals in time and may be approached using the probabilistic models of renewal theory, models which express the concept of 'risk' in terms of characteristics of the distribution of the time to failure. With this approach, Cox (1972) has described a very general regression model for the analysis of survival data and this model may also be applied in the analysis of longitudinal studies of incidence.

Longitudinal studies of incidence and survival studies differ, however, in an important respect in the manner in which the time dimension is interpreted and it is important to clarify this difference before continuing to the main results of this paper. In either type of study consider an individual who enters the study at a specified point in time, i.e. date d_0, say, at which point he is aged s_0, so that his date of birth is $d_0 - s_0$. Both types of study are concerned with the risk of failure of such an individual at some later date, d say, at which time his age is given by $s = s_0 + d - d_0$. Expression of this risk in terms of a failure-time distribution may be approached in two ways depending upon whether the failure time is measured from the date of birth, $d_0 - s_0$, or from the date of entry into the study, d_0. Usually in survival studies, d_0 represents the date at which a patient was diagnosed and the starting date for therapy. Thus, the prime interest is in survival after this date and it is most natural to formulate models with the survival time measured from d_0 as the measure of outcome. The age at diagnosis, s_0, would typically be regarded as a regressor. By contrast, in longitudinal studies of incidence the date d_0 does not correspond to any significant event in the individual's medical history but to the point of time at which the epidemiologist commences observing him. In such studies of previously healthy individuals, the prime interest is in the age at failure so that it is more natural to formulate models with the survival time measured from the date of birth as the measure of outcome. Date of birth itself might be regarded as a regressor so that birth-cohort effects upon outcome may be incorporated into the model.

With this viewpoint, if $f(s|y)$ represents the density of the age at which individuals characterized by the regressor vector, y, succumb to the disease studied and $\mathscr{F}(s|y)$ represents the corresponding survivor function,

$$\mathscr{F}(s|y) = \int_s^\infty f(u|y)\,du,$$

then the age-specific incidence rates for such individuals are given by the hazard function of renewal theory:

$$\lambda(s|y) = \frac{\partial}{\partial s}\{-\log \mathscr{F}(s|y)\} = f(s|y)/\mathscr{F}(s|y).$$

In its simplest form, Cox's model for relating the regressors, y, to survival may be written

$$\lambda(s|y) = a(s)\,e(y). \tag{1}$$

Thus, if the date of birth were the regressor, then the model would hold that the age-specific

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incidence rate may be expressed as a product of a function of age alone and a function of date of birth alone and this model has been commonly used in the analysis of mortality rates (Kermack, McKendrick & McKinlay, 1934). Secular influences operating at the time of failure (Sacher, 1960) could be incorporated by the introduction of an age-dependent regressor term in (1), but such models will not be discussed here.

The model (1) also yields a familiar prediction for the special case of a dichotomous regressor. Such a variable divides individuals into two groups and the model predicts that the ratio between the age-specific incidence rates in these two groups remains constant at all ages, i.e.

$$\lambda(s\,|\,y = 1)/\lambda(s\,|\,y = 0) = e(1)/e(0) = \theta, \tag{2}$$

say. An estimate of the parameter θ could be termed an age-standardized incidence ratio. Kilpatrick (1962) has argued that the proper use of the standardized mortality ratio requires its interpretation as an estimate of the parameter of just such a model.

In general, inference with the model (1) from longitudinal studies of incidence closely follows the argument of Cox (1972). A parametric form, $e(y;\beta)$ say, is assumed for the influences of the regressors and the ageing term, $a(s)$, is regarded as a nuisance function. Inference concerning β from the censored observations in the absence of knowledge of $a(s)$ may be achieved by arguing conditionally upon the ages at which failures occurred in the study. A conditional likelihood is provided by calculating, for each such age, the probability that, of all individuals in the study and still healthy at that age, one particular individual was the one to succumb.

This conditional argument applies virtually unchanged for the analysis of case-control studies in which each case is compared with several age-matched controls. For each set of a case and his controls, given the values of the regressors for each individual, the model predicts the probability that each member is, in fact, the case rather than a control. This provides a likelihood which is independent of the nuisance function $a(s)$.

3. Familial tendency in disease incidence

The work described below arose from the need to find an index to express the influence of parental history of a given disease upon the incidence in the offspring. The difficulty presented by this problem is that we must contend not only with the usual problem of censorship of observations of the major variable of interest, age at incidence of the offspring, but also with censored observations of the regressor, parental history. For example, if a positive parental history of heart disease is defined by a father's succumbing to this disease before age 60 say, then how do we regard fathers who die of cancer at some much earlier age?

Here, this problem is approached by considering for pairs of related individuals, e.g. sons and fathers, the bivariate distribution of age at failure of son, s say, and age at failure of father, t say. The required index is achieved by adopting a suitable single-parameter model for association in this bivariate distribution. The requirements for such a model were considered to be as follows.

(i) It should be relatively straightforward to estimate the association parameter given censored observations of both variables.

(ii) There should be a clear relationship with established models for incidence data. In particular the effect of parental history should be expressible as a constant ratio of age-specific rates as in (2).

(iii) The model should be symmetrical in the two variables.

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The final requirement follows from the type of association considered. It is assumed that association arises because the two members of a pair share some common influence and not because one event influences the other. Thus, if the father's experience predicts the son's then the experience of the son should similarly predict the father's.

Only one model seems to meet these requirements. If a positive history of disease in the father is defined by the father's succumbing at a specified age, $t = t_0$ say, and a negative history by $t > t_0$, then the second requirement suggests that the ratio of age-specific incidence rates for sons, $\lambda_s(s \mid t = t_0)/\lambda_s(s \mid t > t_0)$ should be constant for all s. Likewise, the third requirement suggests that the similar comparison can be made between fathers with and without a history of incidence in sons, so that the ratio of age-specific incidence rates for fathers, $\lambda_f(t \mid s = s_0)/\lambda_f(t \mid s > s_0)$ should be constant for all t. These considerations lead to the model described in the next section and in later sections it is demonstrated that the first requirement is also met by this model; its parameter is easy to estimate using an appropriate conditional likelihood.

4. A BIVARIATE MODEL FOR ORDERED PAIRS

As above, let s represent the age at failure of the first member of an ordered pair of individuals, let t represent the age at failure of the second member and let $f(s,t)$ represent the joint age-at-failure distribution. The model proposed, is that for all s and t

$$f(s,t)\int_s^\infty \int_t^\infty f(u,v)\,dv\,du = \theta \int_s^\infty f(u,t)\,du \int_t^\infty f(s,v)\,dv. \tag{3}$$

In terms of the conditional age-specific rates discussed in § 3, the model may be written that for all s_0 and t_0

$$\frac{\lambda_s(s_0 \mid t = t_0)}{\lambda_s(s_0 \mid t > t_0)} = \frac{\lambda_f(t_0 \mid s = s_0)}{\lambda_f(t_0 \mid s > s_0)} = \theta.$$

It is convenient to adopt a nomenclature drawn from the comparison of disease incidence in fathers and sons; the first member of a pair will be referred to as the son and the second as the father. It is also convenient to define the following four functions derived from $f(s,t)$:

(i) the joint survivor function

$$\mathscr{F}(s,t) = \int_s^\infty \int_t^\infty f(u,v)\,dv\,du;$$

(ii) the hazard function for sons of fathers who survive until t, that is

$$g(s;\,t) = \frac{\partial}{\partial s}\{-\log \mathscr{F}(s,t)\} = \int_t^\infty f(s,v)\,dv/\mathscr{F}(s,t);$$

(iii) the hazard function for fathers of sons who survive until s, that is

$$h(t;\,s) = \frac{\partial}{\partial t}\{-\log \mathscr{F}(s,t)\} = \int_s^\infty f(u,t)\,du/\mathscr{F}(s,t);$$

(iv) the bivariate failure rate (Basu, 1971; Puri & Rubin, 1974)

$$l(s,t) = f(s,t)/\mathscr{F}(s,t) = g(s;\,t)\,h(t;\,s) - \frac{\partial^2}{\partial s\,\partial t}\{-\log \mathscr{F}(s,t)\}.$$

In terms of the earlier notation, $g(s;\,t_0)$ and $h(t;\,s_0)$ correspond to $\lambda_s(s \mid t > t_0)$ and $\lambda_f(t \mid s > s_0)$ respectively, and together correspond to the vector bivariate hazard rate (Johnson & Kotz,

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1975). In terms of these functions, the model (3) may be rewritten as that, for all s and t,

$$l(s,t) = \theta g(s;\, t)\, h(t;\, s). \tag{4}$$

The parameter, θ, of this model measures the degree of association between s and t, independence being implied by $\theta = 1$, and positive association by $\theta > 1$. Inverse associations would be implied by $\theta < 1$ but this case seems to have little practical importance, at least in the present context.

The model may also be expressed in terms of the second-order partial differential equation

$$\frac{\partial^2}{\partial s\, \partial t}\{-\log \mathscr{F}(s,t)\} + (\theta-1)\frac{\partial}{\partial s}\{-\log \mathscr{F}(s,t)\}\frac{\partial}{\partial t}\{-\log \mathscr{F}(s,t)\} = 0,$$

and solution of this equation shows that, when $\theta \neq 1$, the four functions described above must be expressible in the following forms:

$$\mathscr{F}(s,t) = [1+(\theta-1)\{a(s)+b(t)\}]^{-1/(\theta-1)}, \tag{5}$$

$$g(s;\, t) = a'(s)/[1+(\theta-1)\{a(s)+b(t)\}], \tag{6}$$

$$h(t;\, s) = b'(t)/[1+(\theta-1)\{a(s)+b(t)\}], \tag{7}$$

$$l(s,t) = \theta a'(s)\, b'(t)/[1+(\theta-1)\{a(s)+b(t)\}]^2, \tag{8}$$

where $a(.)$, $b(.)$ are nondecreasing functions with $a(0) = b(0) = 0$. It follows that the joint failure-time distribution $f(s,t)$ must be of the form

$$f(s,t) = \theta a'(s)\, b'(t)/[1+(\theta-1)\{a(s)+b(t)\}]^{2+1/(\theta-1)}. \tag{9}$$

Thus, when $\theta > 1$, $g(s;\, t)$ is decreasing with respect to t so that the age-specific rates for sons decrease with increased survival time of their fathers. Similarly, $h(t;\, s)$ decreases with respect to s. When $\theta = 1$ and there is no association, expressions (5) and (9) require modification; thus,

$$\mathscr{F}(s,t) = \exp[-\{a(s)+b(t)\}], \tag{5a}$$

$$f(s,t) = a'(s)\, b'(t) \exp[-\{a(s)+b(t)\}]. \tag{9a}$$

The same model may be derived rather more directly from Cox's (1972) model. As stated in § 3, it is reasonable to assume that association between the age at failure of son and that of father arises, not because the events influence each other, but because father and son share some unknown factor, either environmental or genetic. Consider a father–son pair who share a common value, x, say, for this unknown factor. Application of Cox's regression model suggests the adoption of hazard functions of the form $a'(s)\, e^{\alpha x}$ and $b'(t)\, e^{\alpha x}$ for son and father respectively, corresponding to survivor functions of $\exp\{-a(s)\, e^{\alpha x}\}$, $\exp\{-b(t)\, e^{\alpha x}\}$. Conditional upon x, failure of father and son may be presumed to be independent so that, as in (9a),

$$f(s,t\,|\,x) = a'(s)\, b'(t)\, e^{2\alpha x} \exp[-e^{\alpha x}\{a(s)+b(t)\}]. \tag{10}$$

If the unknown, x, has a distribution $\psi(x)$ in the population of father–son pairs, then the unconditional joint distribution of failure times is given by

$$f(s,t) = \int_{-\infty}^{+\infty} f(s,t\,|\,x)\, \psi(x)\, dx. \tag{11}$$

A convenient form for $\psi(x)$ is such that $c\, e^{\alpha x}$ has a gamma (η) distribution so that the multipliers $e^{\alpha x}$ have mean η/c and variance η/c^2. With this assumption, from (10) and (11)

$$f(s,t) = \eta(\eta+1)\, c^{\eta}\, a'(s)\, b'(t)\, \{c+a(s)+b(t)\}^{-(2+\eta)}.$$

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With the further restriction $\eta = c$ so that the multipliers $e^{\alpha x}$ have unit mean this becomes

$$f(s,t) = \left(1+\frac{1}{c}\right) a'(s)\, b'(t) \left[1+\frac{1}{c}\{a(s)+b(t)\}\right]^{-(2+c)},$$

and this is identical to (9) with $1/c = (\theta-1)$; $(\theta-1)$, therefore, represents the variance of the multipliers $e^{\alpha x}$ and may be interpreted as a component of variance.

The model described in this section has certain similarities to the class of bivariate distributions discussed by Plackett (1965) and by the author (Clayton, 1974). This would provide a model of the form

$$\int_0^s \int_0^t f(u,v)\, dv\, du \int_s^\infty \int_t^\infty f(u,v)\, dv\, du = \theta \int_0^s \int_t^\infty f(u,v)\, dv\, du \int_s^\infty \int_0^t f(u,v)\, dv\, du,$$

and this is the model which would be suggested by the same considerations as in § 3 if one started from the type of model for location shift in failure-time distributions described by the author elsewhere (Clayton, 1976) rather than from Cox's model. This approach has not been pursued further since the problems of estimation are more difficult and since no inference seems to be possible from case-control studies.

It should, perhaps, be noted that the discussion of bivariate life tables by Cox (1972) bears no direct relationship to the present discussion since it is concerned with the problem of two types of failure within the same individual rather than with the same type of failure in a related pair of individuals.

5. Estimation of the parameter θ from longitudinal studies

Ideally, we require an estimate of the parameter θ but are unwilling to make parametric assumptions concerning the nuisance functions $a(s)$ and $b(t)$. The problem is further complicated by the fact that epidemiological studies allow observation of individuals, both sons and fathers, over only a portion of the age range. These difficulties are readily solved by adopting the appropriate conditional argument. Consider the set, $\mathscr{S}$, of points (s,t) such that $N(s,t)$ complete father and son pairs are observed at (s,t) and such that:

(a) a failure occurs at age s in the son in one and only one of these $N(s,t)$ pairs, and

(b) a failure occurs at age t in the father in one and only one of these $N(s,t)$ pairs.

The inference concerning θ may be made by considering for all such points (s,t) whether the failures of both types occurred in the same father–son pair, $\Delta(s,t) = 1$, say, or in different pairs, $\Delta(s,t) = 0$, say. Since it is easily shown that

$$\operatorname{pr}\{\Delta(s,t) = 1 \mid (s,t) \in \mathscr{S}\} = \theta/\{N(s,t)-1+\theta\},$$
$$\operatorname{pr}\{\Delta(s,t) = 0 \mid (s,t) \in \mathscr{S}\} = \{N(s,t)-1\}/\{N(s,t)-1+\theta\},$$

the log likelihood is given by

$$L(\theta) = \sum_{(s,t)\in\mathscr{S}} \left[\Delta(s,t)\log\left\{\frac{\theta}{N(s,t)-1+\theta}\right\} + \{1-\Delta(s,t)\}\log\left\{\frac{N(s,t)-1}{N(s,t)-1+\theta}\right\}\right],$$

so that

$$\frac{\partial L(\theta)}{\partial(\log\theta)} = \sum_{(s,t)\in\mathscr{S}} \left\{\Delta(s,t) - \frac{\theta}{N(s,t)-1+\theta}\right\}, \tag{12}$$

$$-\frac{\partial^2 L(\theta)}{\partial(\log\theta)^2} = \sum_{(s,t)\in\mathscr{S}} \frac{\theta\{N(s,t)-1\}}{\{N(s,t)-1+\theta\}^2}. \tag{13}$$

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The maximum likelihood estimate, $\hat{\theta}$ of θ is obtained by solving the equation obtained by equating (12) to zero. This is most easily achieved by solving

$$\hat{\theta} = \sum_{(s,t)\in\mathscr{S}} W(s,t;\hat{\theta})\{N(s,t)-1\}\Delta(s,t) \Big/ \sum_{(s,t)\in\mathscr{S}} W(s,t;\hat{\theta})\{1-\Delta(s,t)\}$$

by iterative refinement of the weights

$$W(s,t;\theta) = 1/\{N(s,t)-1+\theta\}$$

starting with $W(s,t;1) = 1/N(s,t)$. The first stage of the iteration yields a consistent estimate which is locally efficient in the region of $\theta = 1$. A test statistic for the hypothesis of independence is provided by the value of (12) at the point $\theta = 1$, that is

$$u = \sum_{(s,t)\in\mathscr{S}} \left\{\Delta(s,t) - \frac{1}{N(s,t)}\right\}.$$

The exact sampling distributions of $\hat{\theta}$ and u must remain unknown since $N(s,t)$ are random variables whose distribution depend upon the unknown ages at which censoring would have occurred had failure not preceded it. However, asymptotic confidence intervals and an asymptotic significance test may be obtained using the second derivative (13), that is asymptotically

$$\operatorname{var}(\log\hat{\theta}) = \left[\sum_{(s,t)\in\mathscr{S}} \frac{\hat{\theta}\{N(s,t)-1\}}{\{N(s,t)-1+\hat{\theta}\}^2}\right]^{-1},$$

$$\operatorname{var}(u\,|\,\theta = 1) = \sum_{(s,t)\in\mathscr{S}} \frac{N(s,t)-1}{\{N(s,t)\}^2}.$$

6. Discrete time

Although the epidemiological applications discussed here do not involve renewal processes in discrete time, it is of theoretical interest to consider the discrete analogue of the model (3).

If failure in sons occur at any of R trials and failures in fathers occur at any of C trials, then let p_{ij} represent the probability that, in any father–son pair, failure occurs in the son at the ith trial and in the father at the jth trial. Following Cox's (1972) treatment of the discrete case, we replace the constant hazard-ratio model described in § 4 by the constant odds-ratio model:

$$p_{ij} \sum_{u=i+1}^{R} \sum_{v=j+1}^{C} p_{uv} = \theta \sum_{u=i+1}^{R} p_{uj} \sum_{v=j+1}^{C} p_{iv}. \tag{14}$$

For any pair of trials, (i,j), let

(i) N_{ij} represent the number of complete father–son pairs observed,

(ii) A_{ij} represent the number of these pairs in which the son succumbed at the ith trial,

(iii) B_{ij} represent the number of these pairs in which the father succumbed at the jth trial, and

(iv) D_{ij} represent the number of these pairs in which the son and father both succumbed at the ith and jth trials respectively.

The appropriate conditional argument is to consider, for each pair of trials (i,j) $(i = 1,\ldots;R-1;j = 1,\ldots,C-1)$, the distribution of D_{ij} conditional upon N_{ij}, A_{ij}, B_{ij} and θ. This is a noncentral hypergeometric distribution with noncentrality parameter θ so that

$$\operatorname{pr}(D_{ij}\,|\,N_{ij},A_{ij},B_{ij};\theta) \propto \frac{A_{ij}!\,B_{ij}!\,(N_{ij}-A_{ij})!\,(N_{ij}-B_{ij})!\,\theta^{D_{ij}}}{D_{ij}!\,(A_{ij}-D_{ij})!\,(B_{ij}-D_{ij})!\,(N_{ij}-A_{ij}-B_{ij}+D_{ij})!\,N_{ij}!}.$$

Each pair of trials (i,j) contributes independently to the information concerning θ, so that, asymptotically, the problem of estimating θ is identical to that of estimating the common

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odds-ratio in a stack of $(R-1)(C-1)$ 2×2 contingency tables. A locally efficient and convenient estimate is provided by the well-known Mantel–Haenszel pooled odds-ratio statistic (Mantel & Haenszel, 1959) and a chi-squared test on one degree of freedom of the hypothesis $\theta = 1$ is given by Mantel (1963). The results of § 5 turn out to be special cases of these procedures with $A_{ij} = 1$ or 0 and $B_{ij} = 1$ or 0 for all (i, j). As in § 5, only asymptotic results are available because the N_{ij} are not fixed constants but random variables whose distributions are unknown owing to censoring. Maximum likelihood estimation is tedious using the hypergeometic likelihood but in most practical situations, adequate approximations may be obtained by using Stirling's approximation for the larger factorial terms.

The theoretical interest of the discrete case now becomes clear in that it relates the model to standard contingency table analyses and, in so doing, suggests methods of extending the model to more dimensions. For example, an application of trivariate life tables might arise in relating disease incidence in sons to history of disease in mothers and in fathers. In the discrete case, each combination of discrete ages (i, j, k) of son, father and mother respectively gives a 2^3 contingency table. The generalized model would hold that the four association parameters of the log linear model, three first-order terms and one second-order term (Plackett, 1974, Chapter 9) are constant for all (i, j, k). The equivalent model in continuous time is obvious.

The discrete theory does have some practical application when there is some information loss. If, for example, the ages at which individuals enter and leave the study are only recorded as an integer number of years, then the discrete model is useful. Such observations are, however, observations in stratified time rather than in discrete time, although a discrete time analysis should suffice. An analysis in stratified time might be approached by assuming $a'(s)$ and $b'(t)$ to be step functions. The likelihood for inference when a parametric form is adopted for $a(s)$ and $b(t)$ is given in § 8.

7. COVARIATES

It will often be the case that the association between failure times of the two members of a pair should be estimated with some adjustment for known characteristics of these members. For example, stature is known to be related to the incidence of coronary disease and stature is heritable. It is therefore appropriate to consider whether there still exists an association between incidence in father and incidence in son if the effect of height is removed.

In general, consider a vector of covariates, y, relevant to incidence in the son and a vector, z, relevant to incidence in the father. The direct derivation of our model of association from Cox's regression model outlined in the latter part of § 4 indicates that the natural model to adopt in the presence of covariates is given by expressions (5) to (9) with

$$a(s) = a_0(s)\exp(\beta^{\mathrm{T}} y), \quad b(t) = b_0(t)\exp(\gamma^{\mathrm{T}} z),$$

where β and γ are vectors of regression coefficients for the two sets of covariates and y and z are adjusted to have zero means.

If an estimate only of θ is required, then a straightforward method is provided by grouping of the data. If the available father–son pairs are divided into strata such that pairs within strata have similar values for both sets of covariates, then the association parameter is constant at θ for all strata and may be estimated by simple extensions of the results of § 5. However, if there is more than one covariate, the data will become very thinly spread with a considerable loss in efficiency. It would be preferable, therefore, to attempt a simultaneous estimation of the parameters θ, β and γ.

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Unfortunately a conditional argument for simultaneous inference concerning θ, β and γ in the absence of knowledge of the nuisance functions $a_0(s)$ and $b_0(t)$ does not seem to be available in general. If, as in § 5, we argue conditionally upon a failure occurring in a son at age s and in a father at age t then the probability that such failures occur in the ith son and the jth father is given for $(s,t) \in \mathscr{S}$ by

$$p_{ij}(s,t) = \frac{q_i(s,t)\, q_j(s,t)\, \pi_{ij}(s,t)\, \theta^{\delta_{ij}}}{\Sigma_{i,j}\, q_i(s,t)\, q_j(s,t)\, \pi_{ij}(s,t)\, \theta^{\delta_{ij}}},$$

where $\delta_{ij} = 0$ when $i \neq j$ and 1 when $i = j$, $q_i(s,t) = 1$ when the ith father–son pair is observed at (s,t) and 0 otherwise, so that $\Sigma_i\, q_i(s,t) = N(s,t)$, and where

$$\pi_{ij}(s,t) = \frac{\exp(\beta^{\mathrm{T}} y_i + \gamma^{\mathrm{T}} z_j)}{1 + (\theta - 1)\{a_0(s)\exp(\beta^{\mathrm{T}} y_i) + b_0(t)\exp(\gamma^{\mathrm{T}} z_i)\}} \times [1 + (\theta-1)\{a_0(s)\exp(\beta^{\mathrm{T}} y_j) + b_0(t)\exp(\gamma^{\mathrm{T}} z_j)\}]^{-1}.$$

In general, $p_{ij}(s,t)$ depends upon $a_0(s)$ and $b_0(t)$, but in certain circumstances this argument allows an approximate solution. If $\mathscr{F}_i(s,t)$ is the probability of survival of the ith father–son pair until (s,t), then

$$\pi_{ij}(s,t) = \{\exp(\beta^{\mathrm{T}} y_i + \gamma^{\mathrm{T}} z_j)\}\{\mathscr{F}_i(s,t)\,\mathscr{F}_j(s,t)\}^{(\theta-1)},$$

and under certain conditions the second term in this expression may be ignored, so that an approximate conditional likelihood becomes available. These conditions occur

(i) if θ is close to 1,

(ii) if the regression coefficients β and γ are close to zero so that $\mathscr{F}_i(s,t) \simeq \mathscr{F}_j(s,t)$ for all (i,j), or

(iii) if the disease is rare so that all the $\mathscr{F}_i(s,t)$ are close to 1.

Obviously, the first of these conditions holds for the purposes of testing the null hypothesis, $\theta = 1$.

8. Parametric models for $a_0(s)$, $b_0(t)$

In general it seems that the only method of fitting the model when there are covariates involves the assumption of a parametric form for the functions $a_0(s)$ and $b_0(t)$. Spline functions would be possible or alternatively the assumption of power law relationships of the form $a_0(s) \propto (s-k)^r$ would give a bivariate analogue of the Weibull distribution which empirically seems to provide a good description of the age distribution of chronic disease incidence (Doll, 1971). Detailed discussion of such procedures is beyond the scope of this paper, but the likelihood is straightforward. Consider a pair in which the son is observed from age s_0 until age s_1 and the father is observed from age t_0 until age t_1. If d and e are binary variables indicating whether or not, with values of 1 and 0 respectively, the son and the father succumbed to disease at s_1 and t_1 respectively, then the contribution of such a pair to the log likelihood is

$$\begin{aligned} & de \log \frac{f(s_1,t_1)}{\mathscr{F}(s_0,t_0)} + d(1-e)\log\left\{\int_{t_1}^{\infty} f(s_1,v)\,dv \Big/ \mathscr{F}(s_0,t_0)\right\} \\ & \qquad + e(1-d)\log\left\{\int_{s_1}^{\infty} f(u,t_1)\,du \Big/ \mathscr{F}(s_0,t_0)\right\} + (1-e)(1-d)\log\frac{\mathscr{F}(s_1,t_1)}{\mathscr{F}(s_0,t_0)} \\ & = de\log\theta + d\log g(s_1;\, t_1) + e\log h(t_1;\, s_1) + \log\mathscr{F}(s_1,t_1) - \log\mathscr{F}(s_0,t_0). \end{aligned}$$

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9. CASE-CONTROL STUDIES: A NUMERICAL EXAMPLE

An alternative methodology for the investigation of familial tendency in disease incidence is provided by case-control studies. For example, the histories of the fathers of sons who succumb to disease, i.e. cases, may be compared with the histories of the fathers of age-matched controls. If a case who succumbs at age s is compared with N controls who remained free of disease until s, then the hazard function for the fathers of controls is $h(t;\, s)$ and the hazard function for the father of the case is, according to the model (3),

$$f(s,t) \Big/ \int_t^\infty f(s,v)\,dv = \theta h(t;\, s).$$

The estimation of θ may be accomplished by the conditional argument described in § 2; for each case-control set of $(N+1)$ fathers we argue conditionally upon the ages at which any of them succumbed to disease. For each age, t, such that (i) a father of the case or a control succumbed at age t, (ii) the father of the case survived and was observed at least until t and (iii) the fathers of $n > 0$ of the N controls survived and were observed at least until t, then the probability that the disease occurred in the father of the case rather than in the father of a control is $\theta/(n+\theta)$.

Table 1. *A comparison of mortality from coronary attacks of fathers of heart attack cases with fathers of age-matched controls*

Events providing information concerning θ	No. of occurrences	Likelihood
1. Coronary death in father of the case at an age at which fathers of both controls were observed and alive	11	$\dfrac{\theta}{2+\theta}$
2. Coronary death of father of control at an age at which fathers of both the case and the other control were observed and alive	9	$\dfrac{2}{2+\theta}$
3. Coronary death in father of the case at an age at which the father of only one control was observed and alive	10	$\dfrac{\theta}{1+\theta}$
4. Coronary death in father of one control at an age at which only the father of the case was observed and alive	7	$\dfrac{1}{1+\theta}$

Table 1 shows some data drawn from a study carried out by Morris *et al.* (1973) in which 214 heart attack cases were each compared with 2 age-matched controls. The table compares mortality from heart attacks of fathers of cases with fathers of controls and lists the frequencies of occurrence of the four types of event which provide information concerning the association parameter, θ. The log likelihood is

$$21 \log \theta - 17 \log (1+\theta) - 20 \log (2+\theta)$$

and has a maximum at $\theta = 1{\cdot}926$. The likelihood ratio chi-squared test of the null hypothesis, $\theta = 1$, is 3·83, with one degree of freedom, and a chi-squared test based upon the value of the first derivative at $\theta = 1$ gives 3·91. The log likelihood of § 5 is of a similar form so that the calculations for longitudinal studies are very similar to the above, if more extensive.

10. FURTHER PROBLEMS

The models discussed above may be applied to studies of association in unordered pairs. In such studies the two members of a pair are not distinguished from one another except, possibly, by values of covariates. The models discussed above are unchanged except that the necessary symmetry is included by the restriction $a_0(s) = b_0(s)$ and by using a single regression equation for covariates rather than a separate equation applied to each type of

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individual. Clearly, these assumptions are only legitimate if both members of a pair are studied using an identical methodology and such studies are uncommon. Estimation would require a modification of the conditional argument.

This paper falls short of a complete solution to the statistical analysis of epidemiological studies of familial tendency in chronic disease incidence. Nevertheless, the class of models proposed seems to provide a promising approach to such problems.

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